

On Efficient Quantum Block Encoding

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Abstract

1 Notations and preliminaries

1.1 Notations

We adopt the commonly used notation for binary numbers: for an integer power of two $P = 2^p$ and any $y \in [P]$ if $y = (y_{p-1}y_{p-2} \cdots y_0)$ in the binary system, the corresponding quantum state is $|y\rangle \equiv |y_{p-1} \cdots y_1 y_0\rangle$.

This extends to an m -tuple $\mathbf{x} = (x_1, \cdots, x_m)$ with $x_j \in [P]$. The corresponding quantum state is given by $|x_m\rangle \cdots |x_1\rangle$, where $|x_j\rangle = |x_{j,p-1} \cdots x_{j,0}\rangle, \forall j$. For a multivariate function $g : [P]^m \rightarrow \mathbb{R}$, we denote by D_g the diagonal multiplication operator on $\mathbb{C}^{mp}$:

$$|x_m\rangle \cdots |x_1\rangle \mapsto g(x_1, \cdots, x_m) |x_m\rangle \cdots |x_1\rangle. \quad (1)$$

For a vector $\mathbf{v} = (v_1, v_2, \cdots, v_m)$, we denote $\text{diag}(\mathbf{v})$ as the diagonal matrix $\text{diag}(v_1, v_2, \cdots, v_m)$.

1.2 Discretization of pseudo-differential operators

As mentioned in section intro, the PDO considered in this paper is defined for periodic functions on $\Omega = [0, 1]^d$:

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in \mathbb{Z}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} a(\mathbf{x}, \boldsymbol{\xi}) \hat{f}(\boldsymbol{\xi}).$$

In most numerical treatments, the function f is given on a discrete grid

$$X = \left\{ \frac{\mathbf{x}}{P} \equiv \left(\frac{x_1}{P}, \cdots, \frac{x_d}{P} \right) : x_j \in [P] \right\},$$

where $P = 2^p$ is the number of discrete points used for each dimension. Since the space variable takes value on the Cartesian grid X , the frequency domain is discretized correspondingly on $\{-P/2, \cdots, P/2 - 1\}^d$, which leads to the discretized PDO:

$$Af(\mathbf{x}) \equiv \sum_{\boldsymbol{\xi} \in \{-P/2, \cdots, P/2-1\}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} a\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (2)$$

Though the frequency variable $\boldsymbol{\xi}$ is discretized on $\{-P/2, \cdots, P/2 - 1\}^d$ in (2), by the convention of DFT, the frequency $(P/2, \cdots, P - 1)$ is often identified with $(-P/2, \cdots, -1)$, respectively, since P is a period for the frequency variable after DFT. In other words, the discretized PDO can be written as

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \tilde{a}\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d,$$

where

$$\tilde{a}(\mathbf{x}, \boldsymbol{\xi}) \equiv a \left(\mathbf{x}, \boldsymbol{\xi} - P \sum_{\xi_j \geq P/2} \mathbf{e}_j \right),$$

and $\mathbf{e}_j$ is the j -th standard basis vector in $\mathbb{C}^d$. To simplify the notation and avoid repetitive use of P , we further define

$$\check{a}(\mathbf{x}, \boldsymbol{\xi}) \equiv \tilde{a} \left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi} \right),$$

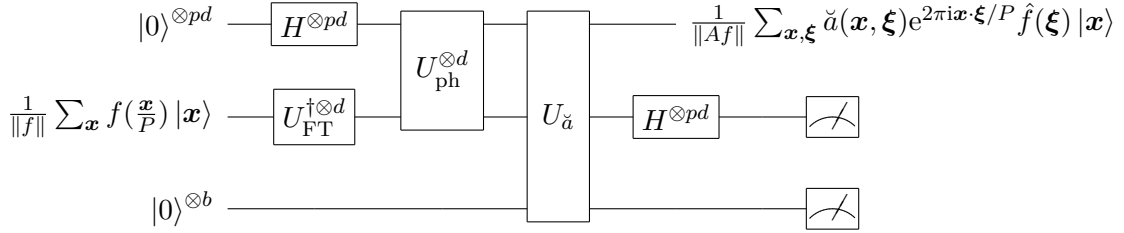
and the discretized PDO becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \check{a}(\mathbf{x}, \boldsymbol{\xi}) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (3)$$

It is clear that $\sup |\check{a}| = \sup |a|$. We also refer to $\check{a}$ as the symbol of the PDO to be computed.

2 Block encoding for generic symbols

This section is concerned with the block encoding of the PDO (3) with a generic symbol $a(\mathbf{x}, \boldsymbol{\xi})$, without assuming any additional structure. In order to compute the PDO in (3), a simple strategy is to first lift the state to the phase space $[P]^d \times [P]^d$. Then the multiplication of $a(\mathbf{x}, \boldsymbol{\xi})$ in (3) can be performed by diagonal matrix block encodings. We can construct the following entire circuit.



Now we explain the above circuit in more details. First, we assume the accessibility of the state $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}>$ as an input, where $\|f\| = \sqrt{\sum_{\mathbf{x} \in [P]^d} |f(\frac{\mathbf{x}}{P})|^2}$ is the normalization factor.

Step 1. Apply QFT and lift the input state to the phase space. We first obtain the representation of f in the frequency domain by QFT. After applying the IQFT to the state $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}>$ for d times, we get

$$\frac{1}{\|f\|} \frac{1}{\sqrt{P^d}} \sum_{\boldsymbol{\xi}} \sum_{\mathbf{x}} f \left(\frac{\mathbf{x}}{P} \right) e^{-2\pi i \boldsymbol{\xi} \cdot \mathbf{x} / P} |\boldsymbol{\xi}> = \frac{\sqrt{P^d}}{\|f\|} \sum_{\boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\boldsymbol{\xi}>. \quad (4)$$

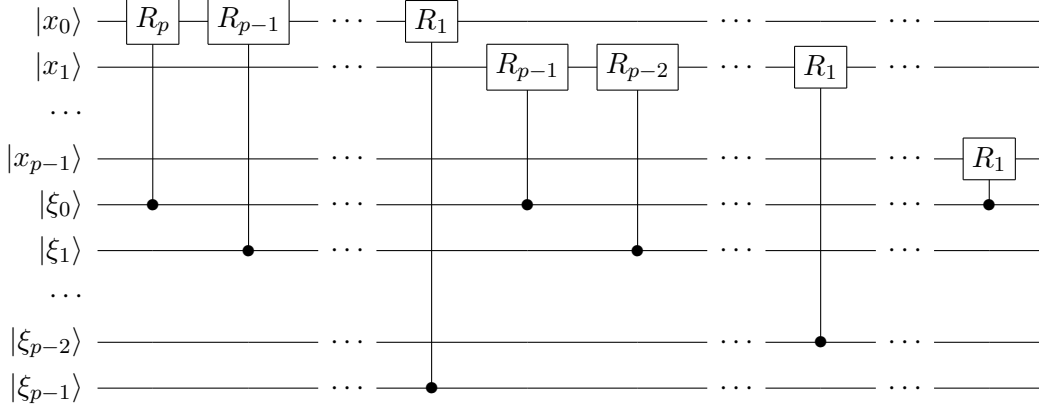
Then the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}> |\boldsymbol{\xi}>$ is obtained by applying the Hadamard gates $H^{\otimes pd}$ to the $\mathbf{x}$ -register and putting both registers together.

Step 2. Multiply the phase $e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P}$ with U_{ph} . We develop an efficient implementation for multiplication without involving any extra error or ancilla qubits in the following lemma.

Lemma 2.1. *The following $2p$ -qubit circuit U_{ph} implements the unitary operator:*

$$|x> |\xi> \mapsto e^{2\pi i x \xi / P} |x> |\xi>, \quad 0 \leq x, \xi < P,$$

with $\mathcal{O}(p^2)$ gate complexity precisely without ancilla qubits.



Proof. We first write the binary representation of integers $x = (x_{p-1} \cdots x_0)$, $\xi = (\xi_{p-1} \cdots \xi_0)$, and do the following calculation:

$$\begin{aligned} e^{2\pi i x \xi / P} |x\rangle |\xi\rangle &= \prod_{j=0}^{p-1} \prod_{k=0}^{p-1} e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_{p-1} \cdots x_0\rangle |\xi_{p-1} \cdots \xi_0\rangle \\ &= \prod_{0 \leq j+k < p} e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_{p-1} \cdots x_0\rangle |\xi_{p-1} \cdots \xi_0\rangle \\ &= \prod_{k=0}^{p-1} e^{2\pi i x_0 \xi_k \cdot 2^{k-p}} |x_0\rangle \otimes \cdots \otimes \prod_{k=0}^0 e^{2\pi i x_{p-1} \xi_k \cdot 2^{k-1}} |x_{p-1}\rangle \otimes |\xi_{p-1} \cdots \xi_0\rangle. \end{aligned}$$

The circuit corresponding to the unitary can be implemented by a series of controlled rotations

$$\begin{array}{ccc} |x_j\rangle & \xrightarrow{R_{p-j-k}} & e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_j\rangle \\ | \xi_k \rangle & \xrightarrow{\bullet} & | \xi_k \rangle \end{array}$$

where $R_j = R_z(\pi/2^{j-1})$. Finally, the above circuit is obtained after arranging the controlled rotations in the corresponding places. $\square$

Rewriting the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ as

$$\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i x_d \xi_d / P} \cdots e^{2\pi i x_1 \xi_1 / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle,$$

then the map from $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ to $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ can be performed by applying [Lemma 2.1](#) for d times to the register pairs $(x_d, \xi_d), \dots, (x_1, \xi_1)$. The corresponding circuit is denoted as $U_{\text{ph}}^{\otimes d}$ and involves $\mathcal{O}(p^2 d)$ elementary gates with no ancilla qubits.

Step 3. Multiply the symbol $\check{a}(\mathbf{x}, \boldsymbol{\xi})$. The next component is the diagonal multiplication $U_{\check{a}}$, which is designed to approximate the map

$$\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b} \mapsto \frac{1}{C_a \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b} + |\perp\rangle, \quad (5)$$

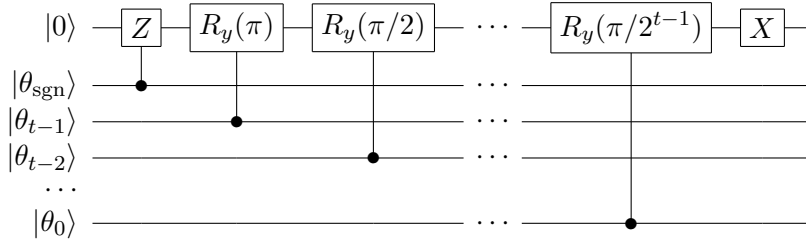
where $C_a > 0$ is a constant that depends on $\check{a}(\mathbf{x}, \boldsymbol{\xi})$ and b is the number of ancilla qubits used for $U_{\check{a}}$. As mentioned earlier, the idea is to treat $(\mathbf{x}, \boldsymbol{\xi})$ as a $2d$ -dimensional variable and utilize the result from arithmetic circuit construction. We state a general result for an efficient block encoding of diagonal matrices D_g , as defined in (1), which is summarized in [Proposition 2.2](#).

Proposition 2.2. Assume that $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is a smooth arithmetic function with $\sup|g| < +\infty$. Then there is an $(C, \mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(mp)), \varepsilon)$ -block-encoding of D_g with $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(mp))$ gates, where D_g is a diagonal operator on $\mathbb{C}^{mp}$ defined in (1), and $C \geq \sup|g|$.

Proof. The circuit U_g is constructed as follows. Let $t = \lceil \log_2(\frac{C\pi}{\varepsilon}) \rceil$, and let $\theta(x_1, \dots, x_m)$ be a map that gives $|\theta_{\text{sgn}}\theta_{t-1}\theta_{t-2}\dots\theta_0\rangle$, where $(\theta_{t-1}\theta_{t-2}\dots\theta_0)$ is the closest t -bit fixed-point representation of $\frac{1}{\pi} \arcsin(|g(x_1, \dots, x_m)|/C)$, and $\theta_{\text{sgn}} = \mathbf{1}_{g < 0}$. For an arbitrary basis state $|x_m\rangle \dots |x_1\rangle$, we consider the system with $t+2$ ancilla qubits $|0\rangle |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1}$. Using the reversible computational model and uncomputation, the classical circuit:

$$|0\rangle |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1} \mapsto |0\rangle |x_m\rangle \dots |x_1\rangle |\theta_{\text{sgn}}\theta_{t-1}\theta_{t-2}\dots\theta_0\rangle$$

can be constructed with $\mathcal{O}(\text{poly}(t) + \text{poly}(mp))$ gates and $\mathcal{O}(\text{poly}(mp))$ ancilla qubits. Then we apply the following circuit on the $t+2$ ancilla qubits.



The state obtained is:

$$(-1)^{\theta_{\text{sgn}}} (\cos(\pi(\theta_{t-1}\dots\theta_0)) |1\rangle + \sin(\pi(\theta_{t-1}\dots\theta_0)) |0\rangle) |x_m\rangle \dots |x_1\rangle |\theta_{\text{sgn}}\theta_{t-1}\theta_{t-2}\dots\theta_0\rangle,$$

which can then be mapped to

$$(-1)^{\theta_{\text{sgn}}} (\cos(\pi(\theta_{t-1}\dots\theta_0)) |1\rangle + \sin(\pi(\theta_{t-1}\dots\theta_0)) |0\rangle) |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1},$$

via uncomputation.

Since

$$\left| (\theta_{t-1}\dots\theta_0) - \frac{1}{\pi} \arcsin(|g(x_1, \dots, x_m)|/C) \right| < \frac{\varepsilon}{C\pi},$$

we have

$$\left| (-1)^{\theta_{\text{sgn}}} \sin(\pi(\theta_{t-1}\dots\theta_0)) - \frac{1}{C} g(x_1, \dots, x_m) \right| < \frac{\varepsilon}{C}$$

which means the desired state $\frac{1}{C} g(x_1, \dots, x_m) |x_m\rangle \dots |x_1\rangle$ is obtained with error at most $\frac{\varepsilon}{C}$ upon measuring the ancilla qubits and getting $|0\rangle^{\otimes t+2}$. $\square$

As a result of applying the block encoding given in Proposition 2.2 with $m = 2d$, $g = \check{a}$ and ε replaced by $\frac{\varepsilon}{\sqrt{Pd}}$ to the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b}$, one obtains the state $|\phi\rangle |0\rangle^{\otimes b} + |\perp\rangle$ using $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(pd))$ ancilla qubits and $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(pd))$ gates where

$$\left\| |\phi\rangle - \frac{1}{C_a \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle \right\| < \frac{\varepsilon}{\sqrt{Pd} C_a}, \quad (6)$$

and $C_a \geq \sup|a| = \sup|\check{a}|$ is a constant. Here $|\phi\rangle$ denotes an unnormalized $2pd$ -qubit state.

Step 4. Sum over the frequency variable. Finally, after applying the Hadamard gate $H^{\otimes pd}$ to the ξ registers, we obtain the state

$$(I_{pd} \otimes |0\rangle^{\otimes pd})(I_{pd} \otimes \langle 0|^{\otimes pd})(I_{pd} \otimes H^{\otimes pd})|\phi\rangle|0\rangle^{\otimes b} + |\perp'\rangle,$$

where

$$\left\| (I_{pd} \otimes |0\rangle^{\otimes pd})(I_{pd} \otimes \langle 0|^{\otimes pd})(I_{pd} \otimes H^{\otimes pd})|\phi\rangle|0\rangle^{\otimes b} \right. \quad (7)$$

$$\left. - \frac{1}{\sqrt{P^d}\|f\|C_a} \sum_{\mathbf{x}, \xi} \check{a}(\mathbf{x}, \xi) e^{2\pi i \mathbf{x} \cdot \xi / P} \hat{f}(\xi) |\mathbf{x}\rangle |0\rangle^{\otimes pd} |0\rangle^{\otimes b} \right\| < \frac{\varepsilon}{\sqrt{P^d}C_a}, \quad (8)$$

which can be seen from (6). Therefore, one obtains the desired state on the $\mathbf{x}$ registers upon measuring $|0\rangle^{\otimes pd}$ for the ξ registers and $|0\rangle^{\otimes b}$ for the ancilla qubits. Notice that there is an extra scaling factor $\frac{1}{\sqrt{P^d}}$ due to the application of Hadamard gates, so the success probability is $\mathcal{O}(2^{-pd})$.

The block encoding scheme of the PDO (3) constructed in this section can be summarized in the following theorem:

Theorem 2.3. *For a generic symbol $a(\mathbf{x}, \xi)$, a block encoding of the corresponding discretized PDO defined by (3) can be $(2^{\frac{pd}{2}}C_a, \mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon})), \varepsilon)$ -block-encoded using the above circuit with gate complexity $\mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$, where $C_a \geq \sup|a|$ is a constant.*

3 Efficient block encoding for separable symbols

In this section, we are concerned with symbols with particular structures and an efficient block encoding of the corresponding PDOs with $\mathcal{O}(1)$ success probability is constructed. We first give the following definition for separable symbols.

Definition 3.1. A symbol $a(\mathbf{x}, \xi)$ is **separable** if $a(\mathbf{x}, \xi) = \alpha(\mathbf{x})\beta(\xi)$.

As explained in Section 1.2, we identify the frequency $(P/2, \dots, P-1)$ with $(-P/2, \dots, -1)$, respectively, since P is a period for the frequency variable after DFT. We also define

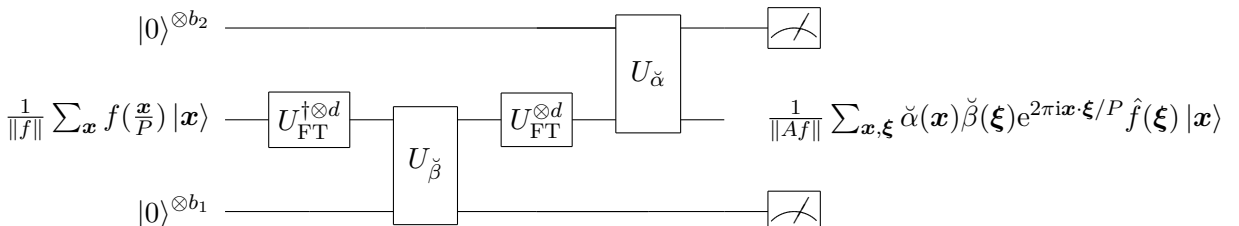
$$\check{\alpha}(\mathbf{x}) = \alpha\left(\frac{\mathbf{x}}{P}\right), \quad \check{\beta}(\xi) = \beta\left(\xi - P \sum_{\xi_j \geq P/2} e_j\right), \quad (9)$$

where e_j is the j -th standard basis vector in $\mathbb{C}^d$. With help of the notations (9), the PDO (2) becomes

$$Af(\mathbf{x}) = \sum_{\xi \in [P]^d} e^{2\pi i \mathbf{x} \cdot \xi / P} \check{\alpha}(\mathbf{x}) \check{\beta}(\xi) \hat{f}(\xi), \quad \mathbf{x} \in [P]^d, \quad (10)$$

It is clear from the definition that $\check{\alpha}$ is P -periodic since α is 1-periodic, and we also have $\sup|\alpha| = \sup|\check{\alpha}|$, $\sup|\beta| = \sup|\check{\beta}|$.

For the PDO (10), we propose an efficient block encoding illustrated by the following circuit.



For the rest of this section, we explain the above circuit with more details.

Step 1. Apply QFT and multiply the factor $\check{\beta}(\xi)$. After applying the QFT gates $U_{\text{FT}}^{\dagger \otimes d}$, one obtains the state $\frac{\sqrt{P^d}}{\|f\|} \sum_{\xi} \hat{f}(\xi) |\xi\rangle |0\rangle^{\otimes b_1+b_2}$. Now that the input is represented on the frequency domain, one can naturally implement the multiplication of the factor $\check{\beta}(\xi)$ since it only depends on the frequency variable ξ . The corresponding block $U_{\check{\beta}}$ can be constructed using [Proposition 2.2](#) with $m = d, g = \check{\beta}$ and ε replaced by $\frac{\varepsilon}{2C_{\alpha}}$, where the constant $C_{\alpha} \geq \sup|\alpha|$. Then one obtains the state $(|\phi_1\rangle |0\rangle^{\otimes b_1} + |\perp_1\rangle) |0\rangle^{\otimes b_2}$, where $|\phi_1\rangle$ satisfies

$$\left\| |\phi_1\rangle - \frac{\sqrt{P^d}}{C_{\beta}\|f\|} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle \right\| < \frac{\varepsilon}{2C_{\alpha}C_{\beta}}, \quad (11)$$

using $\mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$ gates and $b_1 = \mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$ ancilla qubits. Here $C_{\beta} \geq \sup|\beta|$ is a constant.

Step 2. Apply QFT and multiply the factor $\check{\alpha}(x)$. In order to multiply the factor $\check{\alpha}(x)$ in the symbol, we apply QFT and convert the state to the space domain. Since

$$\sqrt{P^d} U_{\text{FT}}^{\otimes d} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle = \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle,$$

we have

$$\left\| U_{\text{FT}}^{\otimes d} |\phi_1\rangle - \frac{1}{C_{\beta}\|f\|} \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \quad (12)$$

$$= \left\| U_{\text{FT}}^{\otimes d} |\phi_1\rangle - U_{\text{FT}}^{\otimes d} \frac{\sqrt{P^d}}{C_{\beta}\|f\|} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle \right\| \quad (13)$$

$$= \left\| |\phi_1\rangle - \frac{\sqrt{P^d}}{C_{\beta}\|f\|} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle \right\| < \frac{\varepsilon}{2C_{\alpha}C_{\beta}}, \quad (14)$$

where we have used (11) in the last line. By using [Proposition 2.2](#) again with $m = d, g = \check{\alpha}$ and ε replaced by $\frac{\varepsilon}{2C_{\beta}}$, the state $|\phi_1\rangle |0\rangle^{\otimes b_2}$ is mapped to $|\phi_2\rangle |0\rangle^{\otimes b_2} + |\perp_2\rangle$, where $|\phi_2\rangle$ satisfies

$$\left\| |\phi_2\rangle - \frac{1}{C_{\alpha}} D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle \right\| < \frac{\varepsilon}{2C_{\alpha}C_{\beta}}.$$

In this step, $\mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$ gates and $b_2 = \mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$ ancilla qubits are used. Therefore, the final state is $|\phi_2\rangle |0\rangle^{\otimes b_1+b_2} + |\perp\rangle$, where $|\phi_2\rangle$ satisfies

$$\begin{aligned} & \left\| C_{\alpha}C_{\beta} |\phi_2\rangle - \frac{1}{\|f\|} \sum_{x, \xi} \check{\alpha}(x) \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \\ & \leq C_{\alpha}C_{\beta} \left\| |\phi_2\rangle - \frac{1}{C_{\alpha}} D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle \right\| + C_{\beta} \left\| D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle - \frac{1}{C_{\beta}\|f\|} \sum_{x, \xi} \check{\alpha}(x) \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \\ & < \frac{\varepsilon}{2} + C_{\beta} \left\| D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle - D_{\check{\alpha}} \frac{1}{C_{\beta}\|f\|} \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \end{aligned}$$

$$\leq \frac{\varepsilon}{2} + C_\beta \sup |\check{\alpha}| \left\| U_{\text{FT}}^{\otimes d} |\phi_1\rangle - \frac{1}{C_\beta \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{\beta}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle \right\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

where we have used the inequality (12) and the fact that $C_\alpha \geq \sup |\alpha| = \sup |\check{\alpha}|$ in the last line. By checking the definition of block encoding and adding up the gates and ancilla qubits used, we obtain the following theorem.

Theorem 3.2. *If $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi})$ is a separable symbol, then the discretized PDO (10) can be $(C_\alpha C_\beta, \mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon})), \varepsilon)$ -block-encoded using the above circuit with gate complexity $\mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$, where $C_\alpha, C_\beta > 0$ are constants such that $C_\alpha \geq \sup |\alpha|$ and $C_\beta \geq \sup |\beta|$.*

Remark. In contrast to the result in Theorem 2.3, one can observe that the exponential factor $2^{\frac{pd}{2}}$ is removed, and thus the success probability for the circuit is improved exponentially compared to the one in Section 2.

4 Efficient block encoding for fully separable symbols with explicit circuits

For separable symbols, the circuit presented in Section 3 significantly increases the success probability compared to the one in Section 2. However, this relies on circuits for arithmetic functions (see Proposition 2.2), which can still be challenging to construct in practice. In this section, we develop a more explicit circuit with the help of QSP and QET.

To begin with, we consider the fully separable symbols defined as follows.

Definition 4.1. A symbol $a(\mathbf{x}, \boldsymbol{\xi})$ is called **fully separable** if $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi})$ with $\alpha(\mathbf{x}) = \alpha_1(x_1) \cdots \alpha_d(x_d)$ and $\beta(\boldsymbol{\xi}) = \beta_1(\xi_1) \cdots \beta_d(\xi_d)$ where each function in the set $\{\alpha_k\}_{1 \leq k \leq d} \cup \{\beta_k\}_{1 \leq k \leq d}$ is a real even function, a real odd function or an exponential function of the form $f(y) = e^{i\theta y}$ for some real parameter θ .

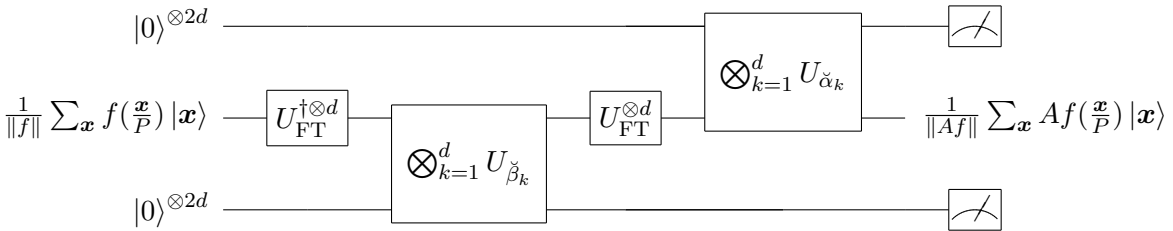
Similar with (9), we introduce the following notations:

$$\check{\alpha}_k(x_k) = \alpha_k\left(\frac{x_k}{P}\right), \quad \check{\beta}_k(\xi_k) = \beta_k(\xi_k - P \cdot \mathbf{1}_{\xi_k \geq P/2}), \quad 1 \leq k \leq d. \quad (15)$$

Then the discretized PDO (2) becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \prod_{k=1}^d \check{\alpha}_k(x_k) \prod_{k=1}^d \check{\beta}_k(\xi_k) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (16)$$

In order to block-encode the PDO (16), we adopt the following circuit, which is similar with the one in Section 3:



Exploiting the fully separable structure of the symbol, one can construct explicit circuits for the diagonal multiplications by leveraging QET. To this end, we first introduce a lemma that gives Hermitian block encodings for two diagonal multiplication prototypes that allow us to build the QET

circuit afterward. More specifically, by combining a series of single-qubit rotations, one can construct a diagonal matrix with diagonal elements $\{e^{ij\theta}\}_{j \in [P]}$. Then one can build a diagonal matrix with diagonal elements $\{\sin(j\theta)\}_{j \in [P]}$ using a simple LCU circuit, from which a diagonal matrix with diagonal elements $\{g(j\theta)\}_{j \in [P]}$ can be built with QET for some smooth function g . Since the grid points of the frequency variables are taken as $\{-P/2, \dots, P/2-1\}$, one also needs the corresponding diagonal matrices where the index j takes values in $\{-P/2, \dots, P/2-1\}$ rather than from 0 to P . We denote the corresponding matrices by subscript $-$ as opposed to $+$ if the index j goes from 0 to P .

Lemma 4.2. For a fixed $P = 2^p$, let $\mathbf{v}_-, \mathbf{v}_+$ be the vectors:

$$\mathbf{v}_- = \left(0, 1, \dots, \frac{P}{2} - 1, -\frac{P}{2}, -\frac{P}{2} + 1, \dots, -1\right), \quad \mathbf{v}_+ = \left(0, 1, \dots, \frac{P}{2} - 1, \frac{P}{2}, \frac{P}{2} + 1, \dots, P - 1\right),$$

respectively, and D_-, D_+ be the diagonal matrices $\text{diag}(\mathbf{v}_-), \text{diag}(\mathbf{v}_+)$, respectively. Then there is an $(1, 1)$ -Hermitian-block-encoding of $\sin(\theta D_\sigma)$ with gate complexity $2p + 5$, where $\sigma \in \{-, +\}$ and $\theta > 0$ is a parameter.

Proof. For an arbitrary parameter $\theta > 0$, we first construct matrices $R_- = e^{i\theta D_-}$ and $R_+ = e^{i\theta D_+}$ with quantum circuits. Using the binary representation of ξ , we get

$$R_- |\xi_{p-1} \dots \xi_0\rangle = e^{i((-\xi_{p-1})\xi_{p-2} \dots \xi_0)\theta} |\xi_{p-1} \dots \xi_0\rangle, \quad R_+ |\xi_{p-1} \dots \xi_0\rangle = e^{i(\xi_{p-1} \dots \xi_0)\theta} |\xi_{p-1} \dots \xi_0\rangle.$$

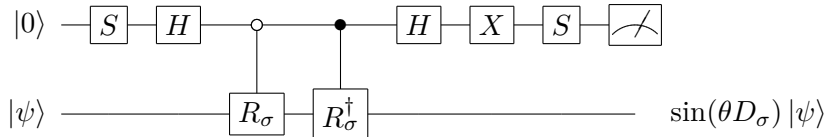
Then

$$\begin{aligned} R_+ |\xi_{p-1} \dots \xi_0\rangle &= e^{i(\xi_{p-1} \dots \xi_0)\theta} |\xi_{p-1} \dots \xi_0\rangle = e^{i \sum_j \xi_j 2^j \theta} |\xi_{p-1} \dots \xi_0\rangle \\ &= \prod_{j \in [P]} e^{i \xi_j 2^j \theta} \bigotimes_{j \in [P]} |\xi_j\rangle = \bigotimes_{j \in [P]} e^{i \xi_j 2^j \theta} |\xi_j\rangle = \bigotimes_{j \in [P]} R_z(2^j \theta) |\xi_j\rangle, \end{aligned}$$

and

$$R_- |\xi_{p-1} \dots \xi_0\rangle = e^{-i \xi_{p-1} 2^{p-1} \theta} R_+ |\xi_{p-1} \dots \xi_0\rangle = \bigotimes_{j \in [P]} R_z((-1)^{\delta_{j,p-1}} 2^j \theta) |\xi_j\rangle,$$

so $R_+ = \bigotimes_{j \in [P]} R_z(2^j \theta)$ and $R_- = \bigotimes_{j \in [P]} R_z((-1)^{\delta_{j,p-1}} 2^j \theta)$. Let U_σ be the following circuit, where $\sigma \in \{-, +\}$, then U_σ is a $(1, 1)$ -Hermitian-block-encoding of $\sin(\theta D_\sigma)$.



In fact, the matrix corresponding to U_σ is

$$\begin{aligned} U_\sigma &= \begin{bmatrix} I & \\ & iI \end{bmatrix} \begin{bmatrix} & I \\ I & \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ I & -I \end{bmatrix} \begin{bmatrix} R_\sigma & \\ & R_\sigma^\dagger \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ I & -I \end{bmatrix} \begin{bmatrix} I & \\ & iI \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} R_\sigma - R_\sigma^\dagger & i(R_\sigma + R_\sigma^\dagger) \\ i(R_\sigma + R_\sigma^\dagger) & -(R_\sigma - R_\sigma^\dagger) \end{bmatrix} = i \begin{bmatrix} \sin(\theta D_\sigma) & \cos(\theta D_\sigma) \\ \cos(\theta D_\sigma) & -\sin(\theta D_\sigma) \end{bmatrix}, \end{aligned}$$

and this is a Hermitian matrix with the first diagonal block being $\sin(\theta D_\sigma)$ after ignoring the global phase factor i . It can be seen that the number of elementary gates used is $2p + 5$. $\square$

Now, we aim to construct the block encoding of diagonal matrices $D_{\check{\alpha}_j} = \check{\alpha}_j(D_+)$ and $D_{\check{\beta}_j} = \check{\beta}_j(D_+)$, namely $U_{\check{\alpha}_k}$ and $U_{\check{\beta}_k}$. For the case where $\alpha_j(x_j) = e^{i\theta x_j}$ or $\beta_j(\xi_j) = e^{i\theta \xi_j}$, the

matrices $R_- = e^{i\theta D_-}$ and $R_+ = e^{i\theta D_+}$ constructed in [Lemma 4.2](#) are exactly the diagonal matrices $D_{\check{\beta}_j} = \beta_j(D_-)$ and $D_{\check{\alpha}_j} = \check{\alpha}_j(D_+)$, respectively. Therefore, we devote the rest of this section to the case where β_j and α_j are even or odd real functions. Thanks to the block encodings U_+ and U_- introduced in [Lemma 4.2](#), what remains to do is to find polynomial approximations of $\check{\alpha}$ and β so as to complete the QET procedure. Specifically, we restrict the parameter θ to be $0 < \theta < \frac{\pi}{2P}$ in order to recover θD_σ from $\sin(\theta D_\sigma)$ with the arcsin function. Going through the QET procedure, one obtains the following result.

Proposition 4.3. *Let U_- and U_+ be the $(1, 1)$ -Hermitian-block-encodings of $\sin(\theta D_-)$ and $\sin(\theta D_+)$ constructed in [Lemma 4.2](#) for $0 < \theta < \frac{\pi}{2P}$ with $P = 2^p$. Assume that g is an even (resp. odd) continuous real function on $[-P, P]$ and that $\deg_g(\varepsilon)$ is the smallest positive integer such that there exists an even (resp. odd) polynomial $\tilde{g}$ with the degree bounded by $\deg_g(\varepsilon)$ satisfying*

$$\sup_{|x| \leq \sin(P\theta)} \left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin x\right) \right| < \varepsilon, \quad \sup_{|x| \leq 1} |\tilde{g}(x)| \leq 1, \quad (17)$$

where $C_g \geq \sup |g|$ is a constant. Then there is a $(C_g, 2, \varepsilon)$ -block-encoding for both $g(D_-)$ and $g(D_+)$ with $\mathcal{O}(p \deg_g(\varepsilon))$ gates, where D_- and D_+ are defined in [Lemma 4.2](#).

Proof. The QET circuit with U_σ gives a $(C_g, 2)$ -block-encoding for $\tilde{g}(\sin(\theta D_\sigma))$, and since

$$\sup_{|x| \leq \sin(P\theta)} \left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin x\right) \right| < \varepsilon,$$

we have

$$\left| C_g \tilde{g}(\sin(\theta D_\sigma)) - g\left(\frac{1}{\theta} \arcsin(\sin(\theta D_\sigma))\right) \right| < \varepsilon,$$

Thus the QET circuit with U_σ gives a $(C_g, 2, \varepsilon)$ -block-encoding for

$$g\left(\frac{1}{\theta} \arcsin(\sin(\theta D_\sigma))\right) = g\left(\frac{1}{\theta} \cdot \theta D_\sigma\right) = g(D_\sigma),$$

where $\sigma \in \{-, +\}$ and we have used the fact that $0 < \theta < \frac{\pi}{2P}$ in the first equality. Since $\mathcal{O}(p)$ gates are used in U_σ , the gate complexity of the circuit described above is $\mathcal{O}(p \deg_g(\varepsilon))$. $\square$

The gate complexity of the circuit built with QET in [Proposition 4.3](#) depends on the smoothness of g . For instance, we have the following corollary.

Corollary 4.4. *Assume that g is an even (resp. odd) differentiable real function on $[-\frac{\pi}{2\theta}, \frac{\pi}{2\theta}]$. Let $\eta_g(\varepsilon, \theta)$ be the smallest integer such that there exists a polynomial u with degree $\eta_g(\varepsilon, \theta)$ satisfying $\sup_{|y| < \frac{\pi}{2\theta}} |u(y) - g(y)| < \frac{\varepsilon}{3}$, then the gate complexity of the circuit used in [Proposition 4.3](#) is $\mathcal{O}\left(p \log\left(\frac{C'_g}{\theta \varepsilon}\right) \eta_g(\varepsilon, \theta)\right)$, where $C'_g = \sup_{|y| < \frac{\pi}{2\theta}} |g'(y)|$.*

Proof. Without loss of generality, assume $1 > \frac{\varepsilon}{2C'_g}$, otherwise we can just let $\tilde{g} = 0$ (17). Since $\frac{1}{\theta} \arcsin(x)$ is an analytic function, there is a truncation v of the Taylor series of $\frac{1}{\theta} \arcsin(x)$ with degree $\mathcal{O}\left(\log\left(\frac{C'_g}{\theta \varepsilon}\right)\right)$ such that $\sup_{|x| < \sin(P\theta)} |v(x) - \frac{1}{\theta} \arcsin(x)| < \frac{\varepsilon}{3C'_g}$. Now since the coefficients of the Taylor series of $\frac{1}{\theta} \arcsin(x)$ at $x = 0$ are all non-negative, it holds that $v([-1, 1]) \subset [-\frac{\pi}{2\theta}, \frac{\pi}{2\theta}]$. Let $\bar{g}(x) = \frac{1}{C_g} u(v(x))$ and $\tilde{g} = (1 - \frac{\varepsilon}{3C_g}) \bar{g}$, then $\tilde{g}$ has degree $\mathcal{O}\left(\log\left(\frac{C'_g}{\theta \varepsilon}\right) \eta_g(\varepsilon, \theta)\right)$, and $\forall |x| \leq \sin(P\theta)$,

$$\left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right| \leq |u(v(x)) - g(v(x))| + \left| g(v(x)) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right|$$

$$< \frac{\varepsilon}{3} + \sup_{|y| < \frac{\pi}{2\theta}} |g'(y)| \left| v(x) - \frac{1}{\theta} \arcsin(x) \right| < \frac{\varepsilon}{3} + C'_g \cdot \frac{\varepsilon}{3C'_g} = \frac{2\varepsilon}{3},$$

In addition, since v maps $[-1, 1]$ into $[-\frac{\pi}{2\theta}, \frac{\pi}{2\theta}]$, we have

$$|\bar{g}(x)| \leq \sup_{|y| < \frac{\pi}{2\theta}} \left| \frac{1}{C_g} u(y) \right| \leq \sup_{|y| < \frac{\pi}{2\theta}} \left| \frac{1}{C_g} u(y) - \frac{1}{C_g} g(y) \right| + \sup_{|y| < \frac{\pi}{2\theta}} \left| \frac{1}{C_g} g(y) \right| \leq \frac{\varepsilon}{3C_g} + 1,$$

and therefore $|\tilde{g}(x)| = (1 - \frac{\varepsilon}{3C_g})|\bar{g}(x)| < 1, \forall x \in [-1, 1]$. In addition, we have $\forall |x| \leq \sin(P\theta)$,

$$\begin{aligned} \left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right| &= \left| \left(1 - \frac{\varepsilon}{3C_g}\right) \left(C_g \bar{g}(x) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right) + \frac{\varepsilon}{3C_g} g\left(\frac{1}{\theta} \arcsin(x)\right) \right| \\ &\leq \left(1 - \frac{\varepsilon}{3C_g}\right) \frac{2\varepsilon}{3} + \frac{\varepsilon}{3C_g} C_g < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon, \end{aligned}$$

According to [Proposition 4.3](#), we have $\deg_g(\varepsilon) = \mathcal{O}\left(\log\left(\frac{C'_g}{\theta\varepsilon}\right) \eta_g(\varepsilon, \theta)\right)$ and the gate complexity of the circuit used in [Proposition 4.3](#) is $\mathcal{O}\left(p \log\left(\frac{C'_g}{\theta\varepsilon}\right) \eta_g(\varepsilon, \theta)\right)$, where the factor p comes from preparing $\sin(\theta D_\sigma)$ as mentioned in [Lemma 4.2](#). $\square$

For the final step, we first introduce the notation

$$\deg_a(\varepsilon) \equiv \sum_{k=1}^d [\deg_{\check{\alpha}_k}(\varepsilon) + \deg_{\check{\beta}_k}(\varepsilon)]. \quad (18)$$

Denote by $U_{\check{\alpha}_k}$ and $U_{\check{\beta}_k}$ the block encodings of $\check{\alpha}_k(D_+)$ and $\check{\beta}_k(D_+) = \beta_k(D_-)$ obtained in [Proposition 4.3](#) with ε replaced by $\varepsilon/2dC$, respectively, where

$$C = \tilde{C}_\alpha \tilde{C}_\beta, \quad \tilde{C}_\alpha = \prod_{k=1}^d C_{\check{\alpha}_k}, \quad \tilde{C}_\beta = \prod_{k=1}^d C_{\check{\beta}_k}, \quad (19)$$

and $C \geq \prod_{k=1}^d (\sup|\check{\alpha}_k| \sup|\check{\beta}_k|)$ since $C_{\check{\alpha}_k} \geq \sup|\check{\alpha}_k|$ and $C_{\check{\beta}_k} \geq \sup|\check{\beta}_k|$. Now we are ready to prove the following theorem that relies on the block encodings $\bigotimes_{k=1}^d U_{\check{\alpha}_k}$ and $\bigotimes_{k=1}^d U_{\check{\beta}_k}$ in the above circuit.

Theorem 4.5. *If $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi}) = \alpha_1(x_1) \cdots \alpha_d(x_d)\beta_1(\xi_1) \cdots \beta_d(\xi_d)$ is a fully separable symbol, then the corresponding PDO defined by (16) can be $(C, \mathcal{O}(d), \varepsilon)$ -block-encoded with gate complexity $\mathcal{O}(dp \deg_a(\frac{\varepsilon}{2dC}) + dp^2)$ using the above circuit.*

Proof. Since each $U_{\check{\alpha}_k}$ is a $(C_{\check{\alpha}_k}, 2, \frac{\varepsilon}{2dC})$ -block-encoding for $\check{\alpha}_k(D_+)$ according to [Proposition 4.3](#), $\bigotimes_{k=1}^d U_{\check{\alpha}_k}$ is a $(\prod_{k=1}^d C_{\check{\alpha}_k}, 2d, \frac{\varepsilon}{2C_\beta})$ -block-encoding for $\bigotimes_{k=1}^d \check{\alpha}_k(D_+) = \bigotimes_{k=1}^d D_{\check{\alpha}_k} = D_{\check{\alpha}}$. Similarly, $\bigotimes_{k=1}^d U_{\check{\beta}_k}$ is a $(\prod_{k=1}^d C_{\check{\beta}_k}, 2d, \frac{\varepsilon}{2C_\alpha})$ -block-encoding for $\bigotimes_{k=1}^d \check{\beta}_k(D_+) = \bigotimes_{k=1}^d D_{\check{\beta}_k} = D_{\check{\beta}}$. Hence by the same argument as the proof of [Theorem 3.2](#), the circuit gives a $(C, 4d, \varepsilon)$ block encoding of the PDO (16) with gate complexity $\mathcal{O}(p \deg_a(\frac{\varepsilon}{2dC}) + p^2 d)$. $\square$